

Markscheme

Mathematics: analysis and approaches

Practice question bank

21. (a)

Consider the function $f(x) = \frac{-2x - 1}{(x + 4)^2}$.

Write $f(x)$ as the sum of partial fractions in the form $\frac{A}{x + 4} + \frac{B}{(x + 4)^2}$.

attempt to write $-2x - 1 = A(x + 4) + B$ (M1)

substituting $x = -4$: $7 = B$ A1

comparing x coefficients (M1)

$-2 = A$ A1

[4 marks]